

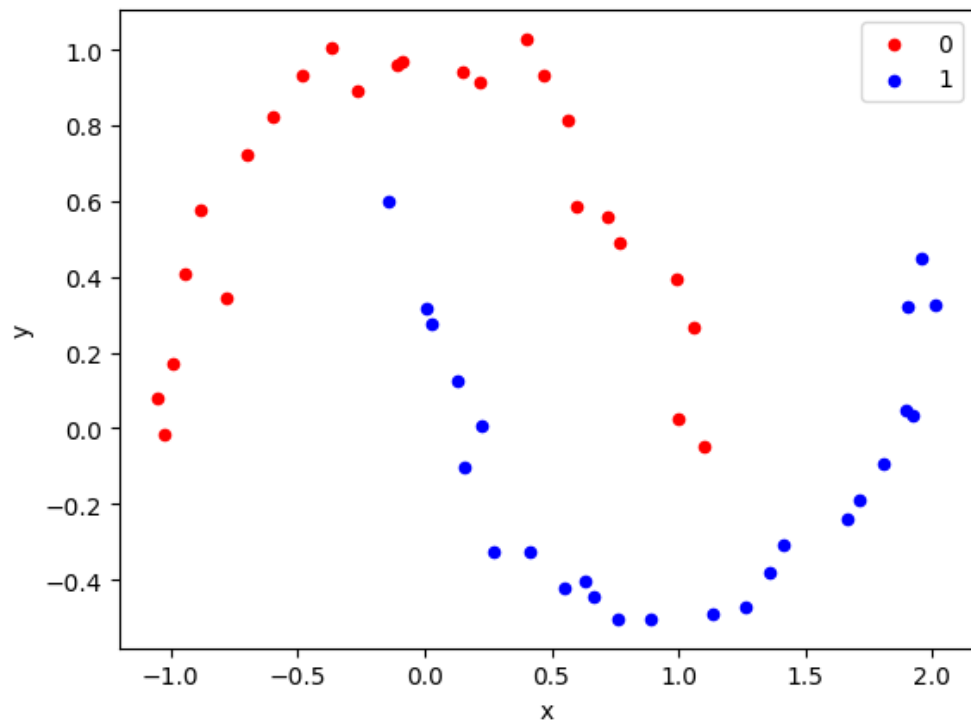
HW2

Classssification

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간단한 모델

```
group.plot(ax=ax, kind='scatter', x='x', y='y', label=key, color=colors[key])  
pyplot.show()
```

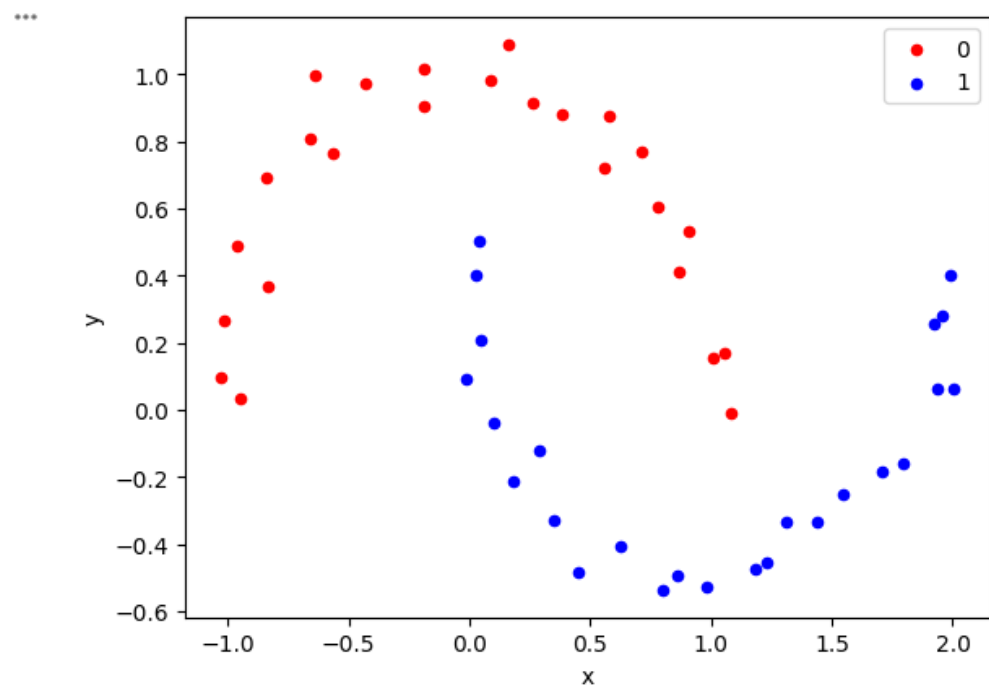


```
model = Sequential()  
# first dense layer  
model.add(Dense(units=2, input_shape=(2,), activation='sigmoid'))  
model.add(Dense(units=4, activation='sigmoid'))  
model.add(Dense(units=4, activation='sigmoid'))  
#model.add(Dense(units=2, activation='sigmoid'))  
model.add(Dense(units=1, activation='sigmoid'))  
# loss function and optimization
```

```
/usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.py  
super().__init__(activity_regularizer=activity_regularizer, **kwargs)
```

중간 정도의 모델

```
# scatter plot, dots colored by class value
df = DataFrame(dict(x=X[:,0], y=X[:,1], label=y))
colors = {0:'red', 1:'blue', 2:'green'}
fig, ax = pyplot.subplots()
grouped = df.groupby('label')
for key, group in grouped:
    group.plot(ax=ax, kind='scatter', x='x', y='y', label=key, color=colors[key])
pyplot.show()
```

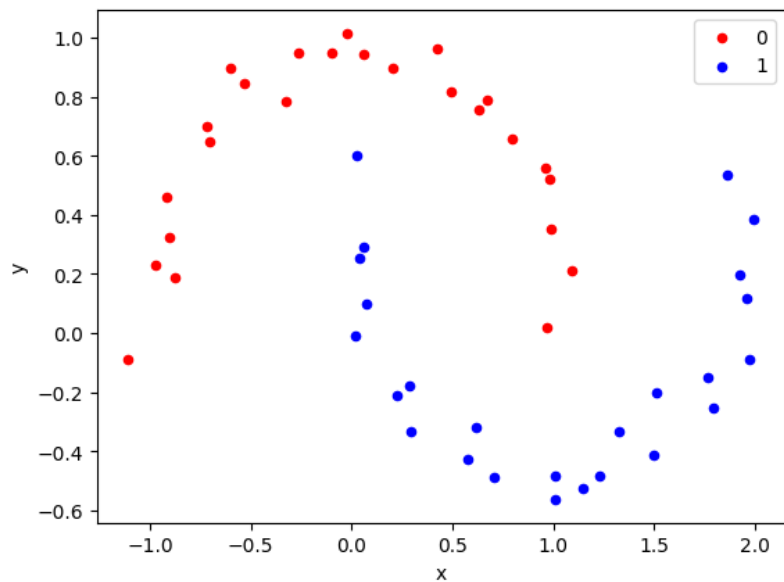


```
> model = Sequential()
# first dense layer
model.add(Dense(units=2, input_shape=(2,), activation='sigmoid'))
model.add(Dense(units=10, activation='sigmoid'))
model.add(Dense(units=10, activation='sigmoid'))
#model.add(Dense(units=2, activation='sigmoid'))
model.add(Dense(units=1, activation='sigmoid'))
# loss function and optimization
```

• /usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.py

가장 복잡한 모델

```
# scatter plot, dots colored by class value
df = DataFrame(dict(x=X[:,0], y=X[:,1], label=y))
colors = {0:'red', 1:'blue', 2:'green'}
fig, ax = pyplot.subplots()
grouped = df.groupby('label')
for key, group in grouped:
    group.plot(ax=ax, kind='scatter', x='x', y='y', label=key, color=colors[key])
pyplot.show()
```



```
model = Sequential()
# first dense layer
model.add(Dense(units=2, input_shape=(2,), activation='sigmoid'))
model.add(Dense(units=8, activation='sigmoid'))
model.add(Dense(units=8, activation='sigmoid'))
#model.add(Dense(units=2, activation='sigmoid'))
model.add(Dense(units=1, activation='sigmoid'))
# loss function and optimization

... /usr/local/lib/python3.13/dist-packages/keras/src/layers/core/dense.
    super().__init__(activity_regularizer=activity_regularizer, **kwar
```